Dayne T. Kovatchev

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Education

University of California, San Diego *Sep 2023 – Present*
B.S. in Electrical Engineering, Power Engineering Specialty (GPA: 3.52) *La Jolla, CA*

- **Active Circuit Design (ECE 102):** Modeled nonlinear circuits (diodes, BJTs, FETs) using small-signal equivalents and simulations to verify lab designs.
- **Devices & Materials (ECE 103):** Studied semiconductor physics, energy bands, and carrier statistics; modeled MOSFET scaling.
- **Electronic Systems Lab (ECE 100):** Designed active op-amp circuits for filter stability and signal conditioning; verified performance using LTspice.
- **Systems Analysis (ECE 101):** Analyzed continuous/discrete LTI systems using Fourier, Laplace, and z-transforms.
- **Electromagnetism (ECE 107):** Explored transmission line electromagnetics, pulse propagation, and high-frequency discontinuities.

Experience

Composite Component Design Engineer Intern (Remote) *Jan 2025 - Jun 2025*
Vescaa.com *Montréal, Canada*

- Designed aerodynamic components (mudflaps, diffusers) for BMW G87 M2 using Fusion 360 forms modeling.
- Developed 3D printed prototypes to ensure sub-millimeter accuracy before carbon fiber manufacturing.
- Collaborated with manufacturers to optimize designs for vacuum infusion and manufacturability.

Component Assembly and Research *Nov 2023 – Feb 2024*
Triton AI *La Jolla, CA*

- Designed precision LiDAR and camera mounts for autonomous scale cars, ensuring rigid sensor alignment.
- Optimized control systems and feedback loops, contributing to a 5% improvement in car centerline response time.
- Conducted hardware testing to validate real-time sensor integration and autonomous steering.
- Managed additive manufacturing workflows using Bambu Lab printers with stringent quality inspection protocols.

Technical Skills

Software: Fusion 360, MATLAB, Python, LTspice, Arduino IDE
Hardware: Oscilloscopes, Function Generators, Multimeters, ESP32, Laser Cutting, LiDAR, Servo Motors
Fabrication: Manual Workshop Tools (Mechanical Fasteners, Woodcutting), 3D Printing
Core Competencies: Active Circuit Design, Signal Processing, Rapid Prototyping, Power Engineering

Projects

Sound-Reactive Mechatronic System *Dec 2025 – Jan 2026*

- Integrated an ESP32 with a MAX9814 sensor to process real-time audio data for complex mechanical response.
- Designed intricate 3D assemblies in Fusion 360 featuring custom wire routing channels and rigid motor mounts for multiple mechanical mechanisms.
- Developed signal processing logic to map audio frequency and amplitude to PWM signals for synchronized motor control.

Leadership

Vice President *Jun 2025 - Present*
UCSD Men's Club Soccer *La Jolla, CA*

- Manage logistics and tournament registration for a 25-player roster while maintaining a full course load.
- Reduced player turnover by fostering a mentorship-driven team culture and improving communication.
- Increased member turnout by establishing consistent social media coverage for team events.